

Experiment 9.1

Discrete Fourier Transform

Program:

```
// Test Case: Discrete Fourier Transform (DFT) of a Discrete-Time Signal
// Define the discrete time signal
x_n = [1, 2, 3, 4];
// Signal x[n]
// Compute the Discrete Fourier Transform (DFT)
X_k = fft(x_n);
// Compute the DFT using Scilab's fft function
// Define the frequency axis
N = length(x_n);
// Number of points in the DFT
k = 0:N-1;
// Frequency index
// Initialize arrays to store magnitude and phase
magnitude = abs(X_k); // Compute magnitude of DFT
phase = zeros(1, N); // Initialize phase array
// Compute phase manually using atan
for i = 1:N
    real_part = real(X_k(i));
    imag_part = imag(X_k(i));
    if real_part > 0 then
        phase(i) = atan(imag_part / real_part); // Phase in [ -pi/2, pi/2 ]
    elseif real_part < 0 then
        phase(i) = atan(imag_part / real_part) + %pi; // Adjust phase for left half-plane
```

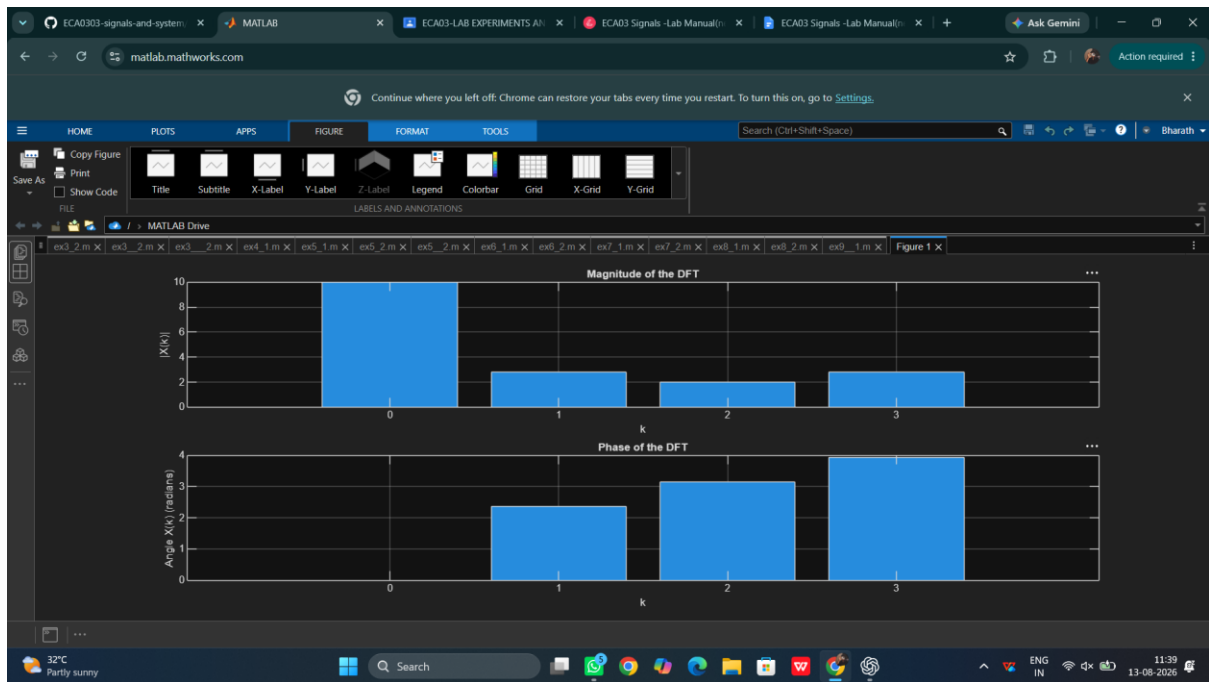
```

else
if imag_part > 0 then
phase(i) = %pi/2; // Positive imaginary axis
elseif imag_part < 0 then
phase(i) = -%pi/2; // Negative imaginary axis
else
phase(i) = 0; // Origin case
end
end
end

// Plot the magnitude and phase of the DFT
clf; // Clear the current figure
// Plot magnitude
subplot(2,1,1);
bar(k, magnitude, 'b');
xlabel('k');
ylabel('|X(k)|');
title('Magnitude of the DFT');
// Plot phase
subplot(2,1,2);
bar(k, phase, 'r');
xlabel('k');
ylabel('Angle(X(k))');
title('Phase of the DFT');

```

Output Waveform:



Result:

This test case will help you compute and visualize the frequency components of a discrete-time signal using the Discrete Fourier Transform in Scilab.